IEEE IAS/IES/PES ESPRIT SBJC PRESENTS

IoT SPARK 4.0

BUILD YOUR EMPIRE



SPECIFICATION BOOK

Seven properties. Seven real problems. One Hardware Bank.

READ ME FIRST

This book explains the technical challenge of IoT Spark 4.0. It has seven properties. Each property is a real engineering problem. At 12:00, your team draws one of them, and the time starts.

HOW TO USE THIS BOOK

- Read the whole book once. Read the Hardware Bank pages and the jury rules with care.
- When you draw your property, read its page again.
- Use the planning sheet at the end of the book during your 45 minutes of planning.
- **Each team receives two links by email. The first link is for your plan, during Build Your Strategy. The second link is for your presentation, at the end of Final Inspection, before you present to the jury.**

WHAT THIS BOOK DOES NOT TELL YOU

It does not tell you which parts to buy, how to build your system, how your devices talk to each other, or which AI to use. Each property gives you a problem, a mission and some limits. You decide how to solve it. The jury will ask you to explain your choices. Two teams with the same property should build two different projects.

THE GAME, STEP BY STEP

Time	Square	What you do
12:00	Pick Your Property	Draw your Property Card and your Constraint Card.
12:10	Build Your Strategy	Plan your solution and list the parts you need. You have 45 minutes. Submit your plan with the first link sent to your team's email.
12:55	Race to the Bank	Buy your parts with your Monopoly credits. The stock is limited.
13:15	Lay the Foundations	Build your hardware.
14:00	Build Your Empire	Write the code, connect the devices, add AI and build your dashboard. You have 3 hours.
17:00	Final Inspection	Test everything and prepare your pitch. A juror visits each team for a short technical check (about 5 minutes). Submit your presentation with the second link sent to your team's email, before you present to the jury.
17:30	Pass GO, Collect Glory	Present and demonstrate your prototype to the jury. You have 8 minutes: 5 minutes of pitch and 3 minutes of questions. Your demonstration must fit the time.

Teams have 2 or 3 members. Before the Race to the Bank, the jury checks your plan and your parts list. If your plan is approved, your team leader can go to the Hardware Bank. All approved teams go at the same time, so it is a real race. A team that arrives late may find some items sold out.

THE THREE EVALUATION MOMENTS

- Plan check: before the Race to the Bank.
- Technical check: during Final Inspection, about 5 minutes per team.
- Final presentation: 5 minutes of pitch and live demonstration, then 3 minutes of questions.

TWO SUBMISSION LINKS

Each team receives two links by email. Link 1: submit your plan during Build Your Strategy, before the plan check. Link 2: submit your presentation at the end of Final Inspection, before you present to the jury.

CARDS AND RULES

Constraint Card. Each team draws one with its property. It adds one rule to your project. The rule is explained when you draw the card. Your plan must follow it.

Change your property. You may re-draw once. A re-draw costs 50 credits, and the new card is final. Two teams may also swap cards for free if both agree. All changes must be done before the plan check.

Chance and Community Chest. During the day, teams draw cards. Some cards bring bad luck. For example, a teammate cannot work for 10 minutes, or you pay 20 credits. Some cards bring good luck. For example, you get 30 extra credits, a free part or one question to a mentor. One card is the jury event: an unannounced test of how your system handles a failure, for example a lost sensor, a fake command or a lost connection.

YOUR BUDGET AND THE HARDWARE BANK

490

MONOPOLY CREDITS FOR EVERY TEAM

Teams of 2 and teams of 3 get the same budget. Every board, sensor and actuator has a price. You can find the full price list on the next two pages. The stock is limited, and all teams share it.

FREE STARTER PACK

Every team gets these for free: one breadboard, 10 male-to-male, 10 male-to-female and 10 female-to-female jumper wires, a set of resistors and USB power at the table. You can buy extra wires in the Hardware Bank.

HOW BUYING WORKS

- Before the Race to the Bank, you submit your plan and your parts list with the first link, and the jury checks them.
- Your parts list is approved when: the planning sheet is complete; the total cost is 490 credits or less (440 after a re-draw); every part is in the price list; the list has at least 2 sensors of different kinds, 1 actuator and 1 board; and your plan says how you protect the system.
- Approved teams buy at the same time. The first to ask is the first served.
- A sold-out part cannot be bought. No part is guaranteed, so prepare a second choice.
- If a part is faulty when you get it, an organizer replaces it.
- Credits you do not spend are not points. The jury scores your choices, not your savings.
- A re-draw of your Property Card costs 50 credits from your budget.

ABOUT THE BOARDS

Two boards are available. Each board has different strengths and limits. Stock is limited, so a part you want may be sold out. The jury scores how well you design with the board you have. It does not score which board you have.

HARDWARE BANK: PRICE LIST (1 OF 2)

Prices are in Monopoly credits. Stock is in units.

BOARDS	Notes	Price	Stock
ESP32 development board	Wi-Fi and Bluetooth	150	14
Arduino Uno	No Wi-Fi	100	5

BREADBOARDS AND WIRES	Notes	Price	Stock
Breadboard, starter pack	One free for each team	Free	11
Jumper wires, male to male	Starter pack: 10 for each team	Free	400
Jumper wires, male to female	Starter pack: 10 for each team	Free	200
Jumper wires, female to female	Starter pack: 10 for each team	Free	200
Extra wires, male to male	Pack of 10	5	10
Extra wires, male to female	Pack of 10	5	10
Extra wires, female to female	Pack of 10	5	10

ENVIRONMENT SENSORS	Notes	Price	Stock
DHT22 sensor	Temperature and humidity	70	8
Gas sensor	Needs warm-up time	90	4
LDR light sensor	Light level	30	8
Water level sensor	Detects water	60	4

MOTION AND POSITION SENSORS	Notes	Price	Stock
Ultrasonic sensor module	Distance	60	6
Motion sensor module	Movement of a person or object	80	5
Infrared obstacle sensor	Short-range detection	40	8
MPU6050	Accelerometer and gyroscope. Costly part	150	2

HARDWARE BANK: PRICE LIST (2 OF 2)

IDENTIFICATION AND INPUT	Notes	Price	Stock
RFID kit	Identification by badge	90	3
Matrix keypad	Key input	50	2
Push buttons DS-316MM	Press to send a signal	10	6
Switch	On/off	10	10

ACTUATORS AND OUTPUTS	Notes	Price	Stock
Buzzer	Sound	20	3
LED, red, green or yellow	One color	10	10 each
1-channel relay module	Switches a larger load	80	8
Servo motor	Moves to a set angle	70	6
Mini water pump	12 V. Low voltage only	100	3
Small fan	Low voltage only	80	3
OLED display	Text and graphics	120	4

POWER AND BUILDING	Notes	Price	Stock
LM2596 step-down regulator	Lowers the voltage	40	9

Resistors and USB power are part of the free starter pack. They are not sold.

THE SIX LAYERS

Every project is built with the same six functional layers. The table shows the minimum for every team. It does not tell you how to build each layer. That is your design work. Your architecture must be coherent across the six layers, and every target of your property must be traceable to a component or a function of your system.

Layer	What we expect from every team
Sensing	At least two sensors of different kinds, relevant to the monitored physical quantity. The readings change consistently with the physical input. One strange reading must not start an alarm, and a faulty or disconnected sensor is detected and handled.
Communication	Data and commands travel between your devices and the operator interface in real time, within the response threshold of your property. The traffic is protected in transit.
Edge processing and AI	A real AI model or service that reads a window of readings. A simple rule such as 'if the value is above...' is not enough. You may use a ready-made model. Follow the AI standard on the next page.
Actuation	At least one actuator that reacts by itself to the detected condition. A person can also control it by hand from the operator interface. It has a defined safe state on failure or on loss of communication.
Security	Protect who can use the system, the data it sends and the commands it receives. Find the risks that matter for your property (at least two realistic threats), and explain how you protect against them with a protection you built.
Operator interface	The dashboard shows live values, a clear status, an alarm indication and an event history. A person can send a command from it. A non-expert operator can read it within a few seconds.

PROTOTYPE AND REAL LIFE

You are building a prototype, not the final product. Think about how the real solution would look. Then explain how your prototype copies the same idea in a smaller and cheaper way: what changes, what is preserved, and why your prototype is still a good model of the real solution. Give an estimate of the unit cost of the real deployment. Each property page gives some real-world facts to help you.

SAFETY FIRST

Use low voltage only. No real fire, flame or smoke. Keep water away from wires and power. If a part breaks, gets hot or smells burnt, stop and call an organizer.

THE AI STANDARD

Every property needs a real AI part. All teams follow the same standard, so the AI work is equal for everyone.

Step	What you do	What you show
1. Use a window	The model reads a window of readings. A window is a short group of readings taken over time, not one single value. The model runs inside your live system, and its answer changes when the input changes.	The window and the answer of the model. The model file.
2. Test it	Test the model on at least 20 labeled windows, both normal and abnormal. A labeled window is an example where you know the right answer, for example ‘normal’ or ‘problem’.	Your test data.
3. Count the errors	Make a confusion matrix. This is a table that counts the right and the wrong answers of your model. Then calculate accuracy, precision and recall.	The table and the metrics, and what the errors mean.
4. Choose your key metric	Choose the metric that matters most for your property, and say why. Analyse the failure modes: what a false alarm costs, what a missed event costs, and the limits of your model.	Your key metric and your analysis of false alarms and missed events.
5. Explain it	Say what goes into the model, what comes out, and when it can be wrong. Explain why the model is suited to your property.	Every member can explain it.

GOOD TO KNOW

A simple model that you explain well scores higher than a strong model that you cannot explain. The jury cares more about your understanding than about the numbers. The model can run on the board or on a laptop. Both are scored the same. Comparing your model with a simple threshold rule is a good way to show why you need a model, but it is not a separate scored step.

HOW YOU WILL BE EVALUATED

Every team is evaluated with the same grid, whatever its property. The total is 100 points. Several jurors score each team on their own, and their scores are averaged. The table shows the main areas.

Each team receives two links by email. The first link is for your plan, during Build Your Strategy. The second link is for your presentation, at the end of Final Inspection, before you present to the jury.

Area	Points	What the jury looks at
Plan and design	13	Your understanding of the problem, your architecture, your choice of parts and your budget.
Build and prototype	24	Your sensors and actuators, and the targets of your property. They must work and keep working.
AI	12	A real model, tested and explained, that follows the AI standard.
Security	10	How you protect users, data and commands.
Operator interface	6	Your dashboard: live values, alerts, history and control.
Technical understanding	5	Every member can explain their own part and the limits of the prototype.
Pitch and demonstration	30	Your pitch, your live demonstration, your teamwork and your time management.
TOTAL	100	

BASIC AND ADVANCED

The mission and the targets of your property are required. The advanced objectives on each property page are optional, and they count inside the 100 points. Doing the basics very well is worth more than starting advanced objectives and leaving the basics unfinished.

TECHNICAL CHECK (DURING FINAL INSPECTION)

A juror visits each team for about 5 minutes. Be ready to show your system working, including its security and its AI, and to explain your own wiring, code and design choices.

3 FIRE DETECTION

Property 3 of 7

THE PROBLEM

Most fires start small. The danger is that nobody sees them in time. Early detection gives people time to leave safely.

YOUR MISSION

Find the early signs of a fire, sound a clear alarm and help reduce the danger.

TARGETS AND LIMITS

Target	Requirement
Levels	The system shows three danger levels, for example normal, warning and alarm, on the dashboard and on the board.
Response	The alarm starts within 20 seconds after the test condition is applied.
Reliability	No alarm in normal conditions during the whole demo.
Reset	Only an allowed person can reset the alarm.

IN THE REAL WORLD

A real building has many rooms and strict safety rules. A real system is certified, tested often and must keep working when a part fails.

ADVANCED OBJECTIVES (OPTIONAL)

- Estimate the risk level and how fast it rises
- Detect a faulty sensor
- Watch two rooms on one dashboard

SAFETY RULES

No real fire, flame or smoke is allowed at the event. The organizers will show you safe ways to test.

TEAM PLANNING SHEET

Use this sheet during Build Your Strategy. You have 45 minutes. Write short and clear answers. Submit it with the first link sent to your team's email. The jury checks it at the plan check.

Each team receives two links by email. The first link is for your plan, during Build Your Strategy. The second link is for your presentation, at the end of Final Inspection, before you present to the jury.

Team name:	**Property:**
Members:	**Team leader:**
Constraint card:	

THE PROBLEM IN OUR OWN WORDS

Who is affected? What can fail? What does an incident or a loss cost?

OUR SIX LAYERS

Layer	What we will do	Who is in charge
Sensing		
Communication		
Edge processing and AI		
Actuation		
Security		
Operator interface		

OUR ARCHITECTURE

Sketch the six layers and show where each target of our property is handled (architecture diagram).

TEAM PLANNING SHEET (CONTINUED)

OUR PARTS LIST

Part	How many	Cost (credits)	Why we need it	Alternative if sold out

Total cost: _____ **Our budget:** 490 credits (440 after a re-draw) **Estimated unit cost of the real deployment:** _____

HOW OUR PROTOTYPE MAPS TO THE REAL SOLUTION

What changes, what is preserved, and why the prototype stays representative.

OUR BIGGEST RISK AND OUR PLAN B (ALSO FOR SOLD-OUT PARTS)
